

FMCG ERP SYSTEM

FMCG ERP Intelligence, Analytics & AI Manual

Analytics · Report Builder · AI Assistant · Custom Reports

Audience: Managers, Analysts, Senior Leadership

Version: v2.0 — Post-Recovery Edition

Date: 24 May 2026

Status: Confidential — Internal Use Only

ERP button/link recovery complete. All navigation targets restored and verified. Smoke test: 141/141 routes passed. BVT: 0. Broken action cards: 0.

FMCG ERP — Intelligence, Analytics & AI Manual v2

Version: 2.0 (post-recovery)

Date: 2026-05-24

Audience: Managers, Analysts, Business Intelligence Teams, All Senior Staff

Modules Covered: Analytics · Report Builder · Sales Analytics · Production Analytics · Inventory Analytics · Finance Analytics · AI Assistant · NPD Analytics

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1. Module Overview

What it does: Business intelligence, analytics dashboards, custom report building, and AI-powered insights across all ERP modules. Enables data-driven decision making through interactive charts, KPI monitoring, and natural-language querying.

Who uses it:

- Managing Director / CEO — top-level KPI monitoring
- Finance Manager — financial performance and forecasting
- Sales Manager — revenue tracking and pipeline analysis
- Production Manager — OEE, yield, and cost analytics

- Supply Chain Manager — inventory and procurement analytics
- All Managers — custom report building

When to use it:

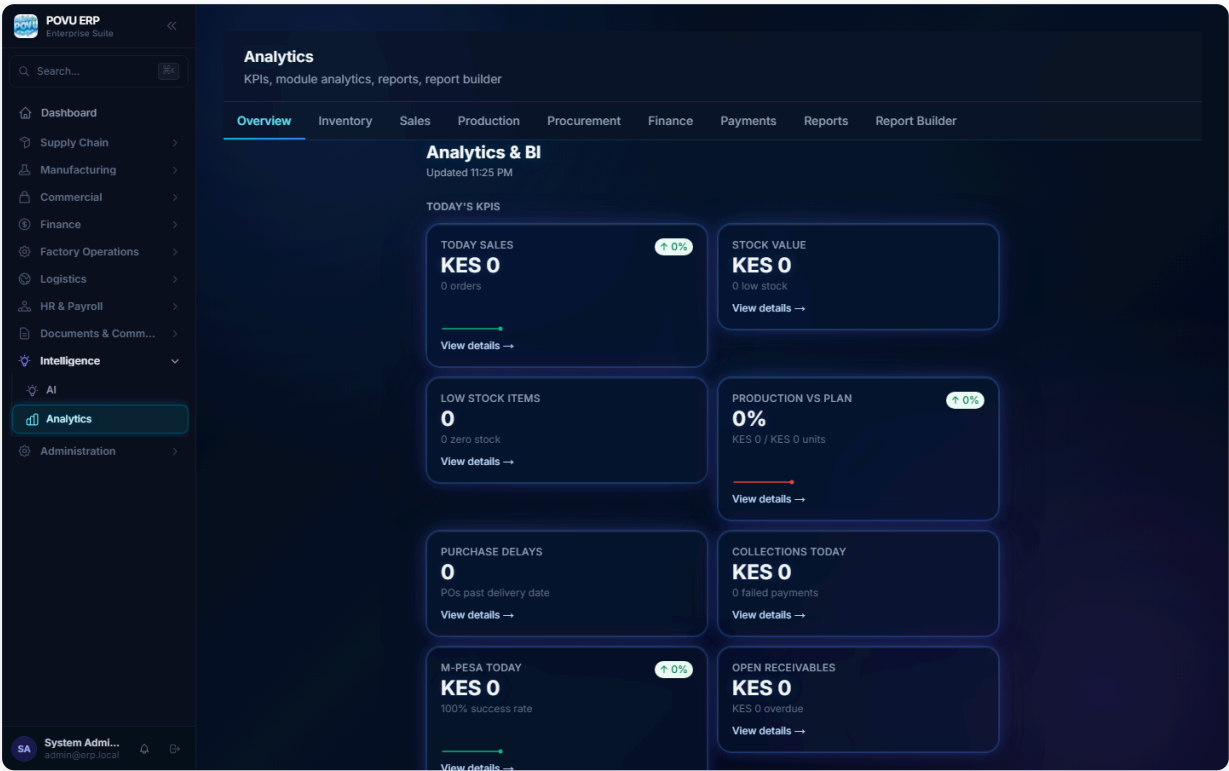
- Daily: Check KPI dashboards at start of business day
- Weekly: Review production OEE, sales performance, inventory levels
- Monthly: Run financial reports for management meetings
- Ad-hoc: Build custom reports for specific analysis needs
- Always available: AI chat for on-demand data queries

Modules at a glance:

Feature	Route	Purpose
Analytics	/dashboard/analytics	Full analytics workspace
AI	/dashboard/ai	AI assistant and insights

2. Analytics Hub

Route: </dashboard/analytics>



Analytics hub — module selector and top-level business KPIs.

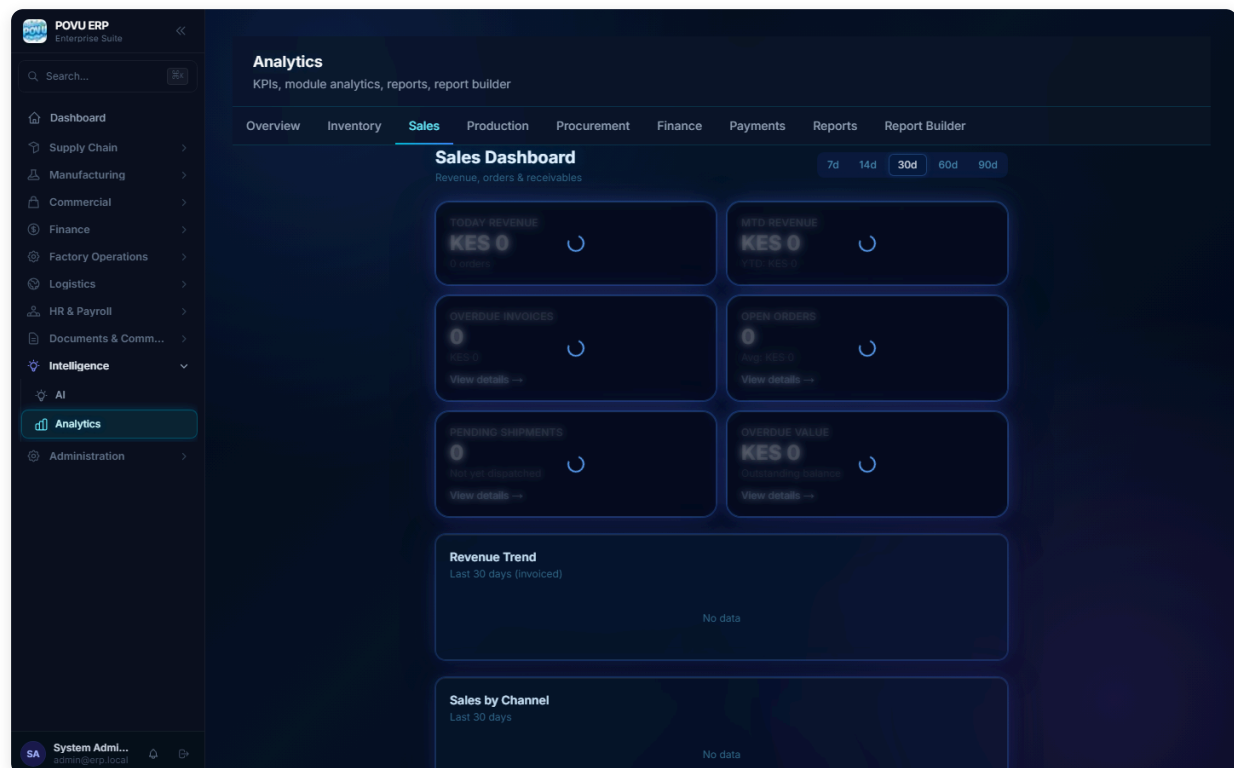
Tabs

Tab	Purpose
Overview	Company-wide KPI dashboard
Sales	Revenue, orders, and customer analytics
Production	Manufacturing performance analytics
Inventory	Stock levels, movements, and aging
Finance	Financial performance and variance
Report Builder	Custom report creation tool

3. Sales Analytics

Tab: Sales

Route: </dashboard/analytics/sales>



Sales analytics — revenue trends, top products, customer rankings, and channel performance.

Metrics Available

Revenue:

- Total revenue (period selectable)

- Revenue vs. same period last year
- Revenue by product category
- Revenue by customer
- Revenue by territory / region
- Revenue by channel (direct / distributor / van / POS)

Orders:

- Order count and average order value
- Order fill rate (% delivered on time in full — OTIF)
- Orders by status
- Order cycle time (order to delivery)

Customers:

- Top 10 customers by revenue
- New vs. returning customers
- Customer lifetime value
- Churn indicators

Products:

- Top selling SKUs by volume and value
- Slow-moving products
- Margin by product
- Price compliance (actual vs. price list)

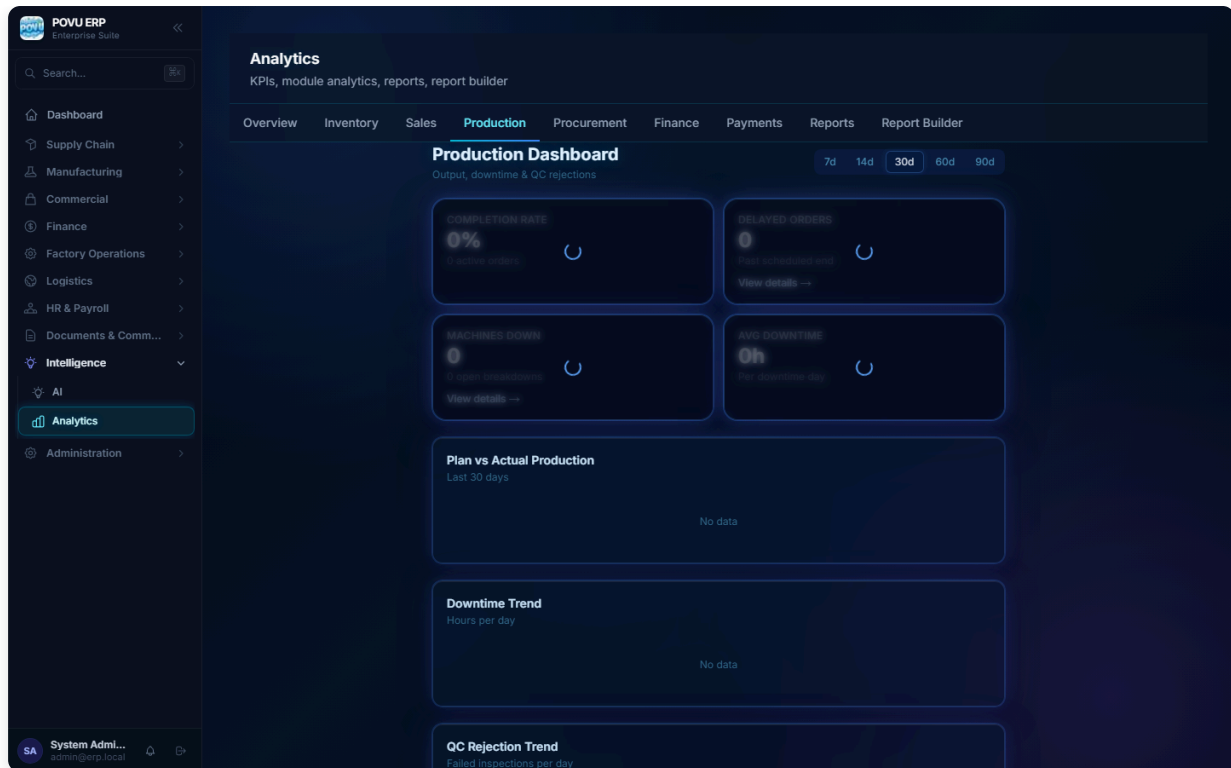
Filters Available

- Date range (custom or presets: Today / This Week / This Month / Last Month / YTD / Last Year)
- Customer
- Product category
- Territory
- Sales rep
- Channel

4. Production Analytics

Tab: Production

Route: </dashboard/analytics/production>



Production analytics — OEE, yield, work order performance, and cost variance.

Metrics Available

OEE:

- Overall OEE % by line and period
- Availability, Performance, Quality breakdown
- OEE trend chart
- Benchmark vs. target OEE

Production Output:

- Planned vs. actual production quantity
- Production plan adherence %
- Batch count and average batch size
- Work order completion rate

Yield & Waste:

- Yield % by product
- Waste quantity and cost
- Scrap categories breakdown
- Yield trend over time

Costing:

- Standard vs. actual cost per unit

- Material variance
- Labor variance
- Total production cost per period

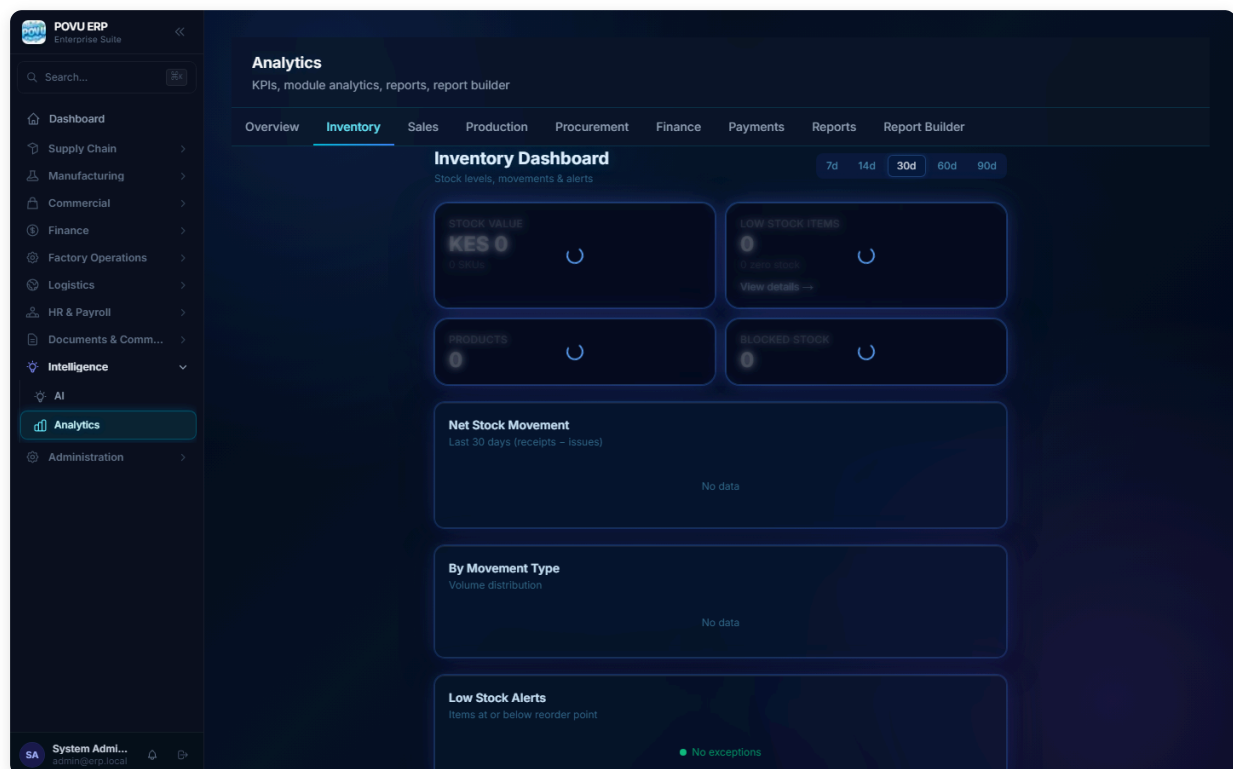
Downtime:

- Total downtime hours
- Downtime by machine
- Downtime by category (Mechanical / Electrical / etc.)
- MTBF and MTTR by asset

5. Inventory Analytics

Tab: Inventory

Route: </dashboard/analytics/inventory>



Inventory analytics — stock levels, turnover, aging, and procurement performance.

Metrics Available

Stock Levels:

- Total stock value by category
- Stock below reorder level (count and list)
- Overstock items (above max stock level)

- Zero-stock finished goods with pending orders

Stock Turnover:

- Inventory days on hand
- Stock turnover ratio by category
- Slow-moving and dead stock
- Shelf life utilization (perishables)

Procurement Performance:

- Supplier on-time delivery %
- PO lead time actuals vs. expected
- Purchase price variance (actual vs. standard)
- Supplier quality score

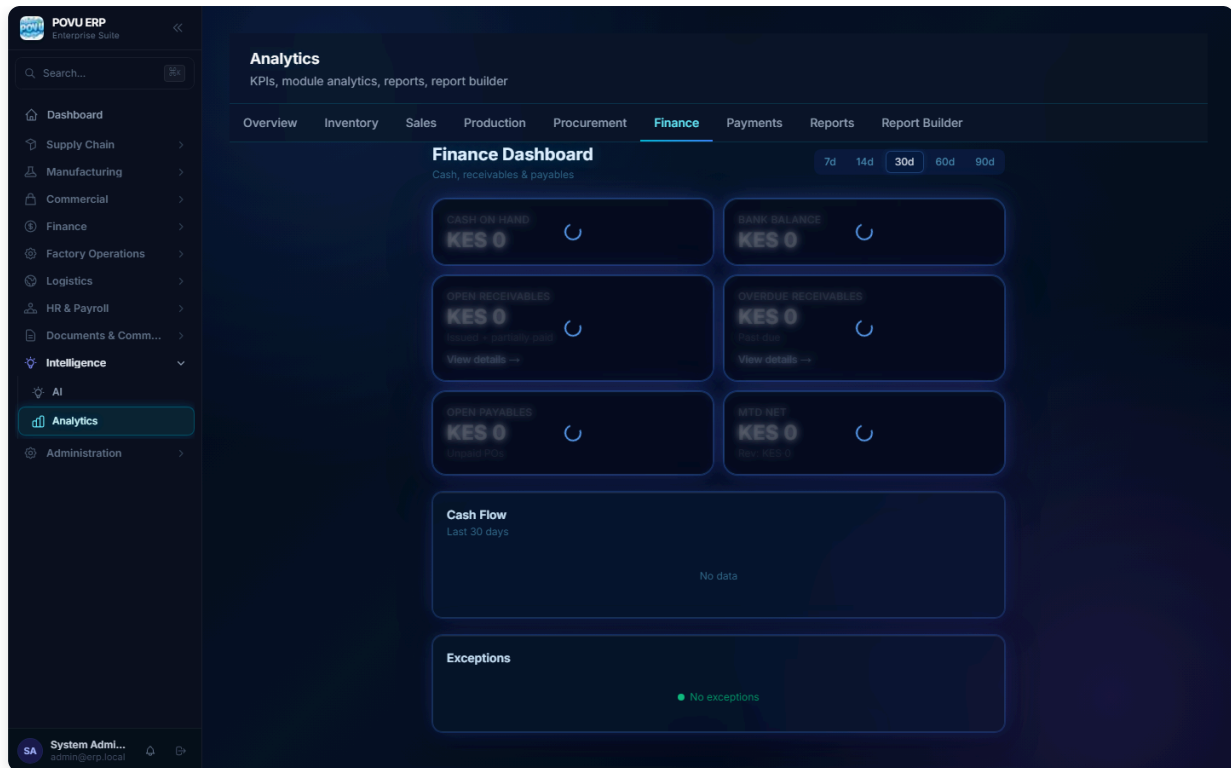
Warehouse:

- Location utilization %
- Picking accuracy
- Goods receipt to putaway time

6. Finance Analytics

Tab: Finance

Route: </dashboard/analytics/finance>



Finance analytics — P&L, cash flow, AR aging, and budget variance.

Metrics Available

P&L Summary:

- Revenue, Cost of Sales, Gross Margin
- Operating Expenses
- EBITDA
- Net Profit
- Period vs. period comparison

Cash Flow:

- Cash inflows (customer collections)
- Cash outflows (supplier payments, payroll)
- Net cash position
- Cash flow forecast (30/60/90 days)

Accounts Receivable:

- Total outstanding AR
- Aging breakdown (Current / 30 / 60 / 90+ days)
- DSO (Days Sales Outstanding)
- Overdue customers list

Budget:

- Budget vs. actual by account and cost centre
- Over-budget items flagged
- Full-year forecast vs. budget

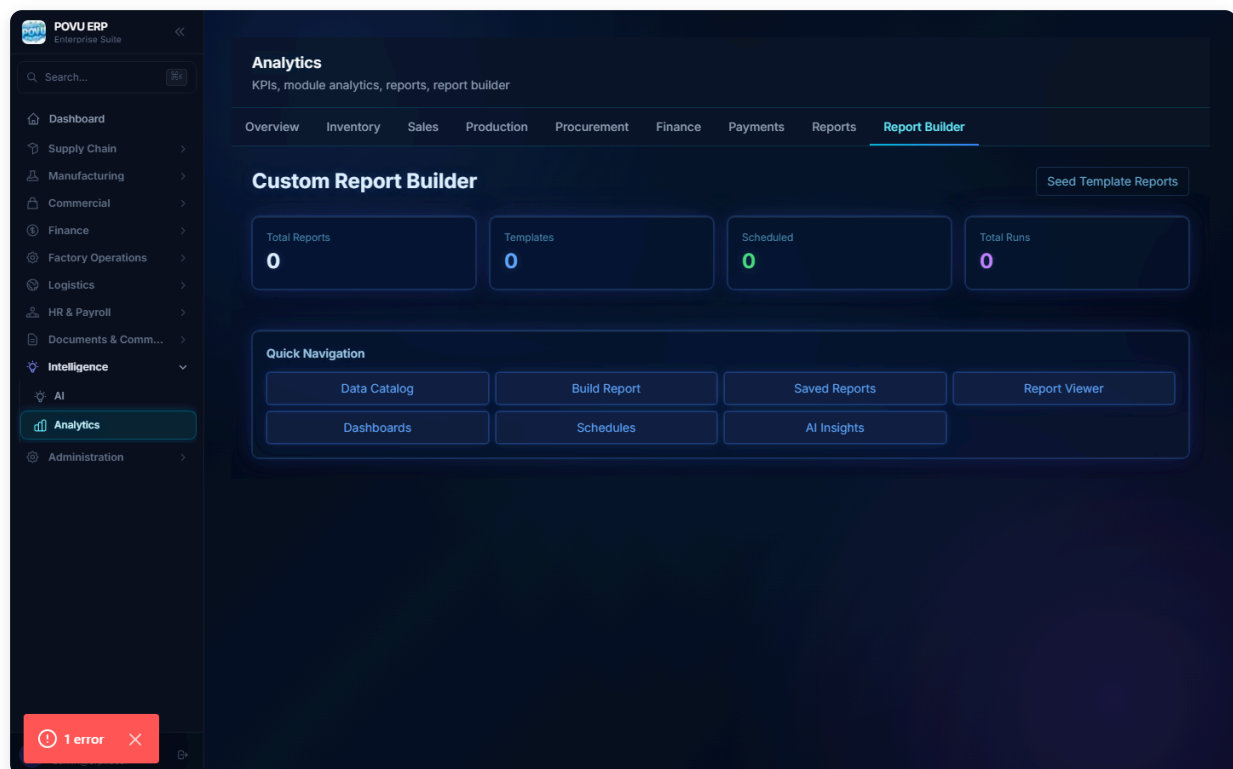
Tax:

- VAT output vs. input summary
- PAYE liability
- WHT payable

7. Report Builder

Tab: Report Builder

Route: `/dashboard/analytics/report-builder`



Report builder — drag-and-drop custom report creation from any ERP data source.

What it does

Build custom tabular and chart reports without needing IT support. Choose data sources, columns, filters, grouping, and visualization type.

Creating a Custom Report

1. Report Builder → + New Report
2. Select **Data Source** (which module/table):

- Sales Orders
- Invoices
- Products
- Purchase Orders
- Inventory Stock
- Production Work Orders
- HR Employees
- Finance Journals
- etc.

3. **Select Columns** — drag fields to include
4. **Add Filters** — optional conditions (e.g. Date > 2026-01-01, Status = COMPLETED)
5. **Group By** — aggregate data (e.g. by Product Category, by Month)
6. **Sort** — set column sort order
7. **Visualization** — Table / Bar Chart / Line Chart / Pie Chart / KPI Card
8. **Save Report** — assign name, category, and access level

Scheduling Reports

After saving:

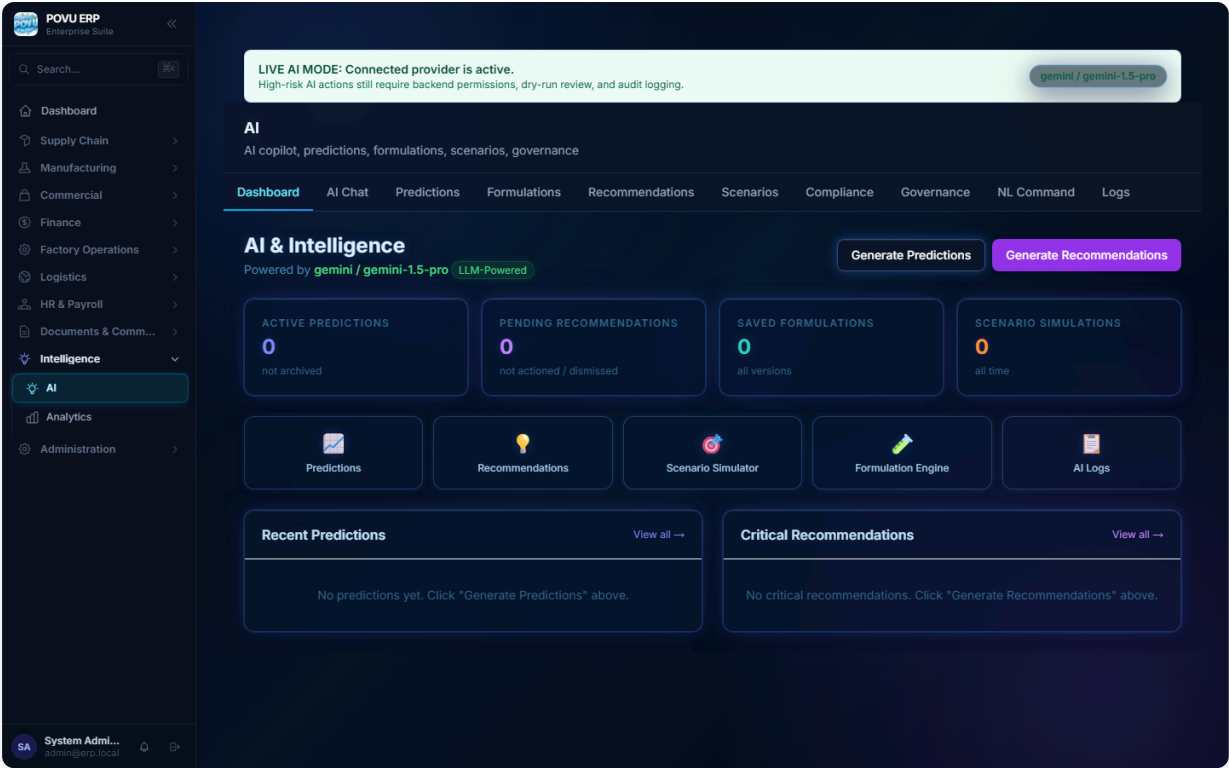
1. Open report → **Schedule**
2. Set frequency: Daily / Weekly / Monthly
3. Set recipients (email addresses)
4. Set format: PDF / Excel / CSV
5. Save — report auto-sent on schedule

Sharing Reports

- Save report as **Public** — all users can view
- Save as **Department** — department members only
- Save as **Private** — only you

8. AI Assistant

Route: `/dashboard/ai`



AI assistant — intelligent recommendations across all ERP modules.

AI Features

Feature	Where Used	What It Does
Demand Forecast	Sales / Inventory	Predicts next 30/60/90 day demand per SKU
Reorder Suggestions	Procurement	Suggests when and how much to reorder
Production Optimization	Manufacturing	Suggests optimal batch sizes and schedules
Quality Anomaly Detection	Quality	Flags unusual QC results for investigation
Financial Insights	Finance	Auto-generates commentary on P&L variances
Customer Insights	CRM / Sales	Identifies at-risk customers and upsell opportunities
Maintenance Prediction	Maintenance	Predicts equipment failures from sensor data

Demand Forecast

The AI demand forecast model:

- 1. Analyzes 24 months of sales history
- 2. Accounts for seasonality, promotions, and trends

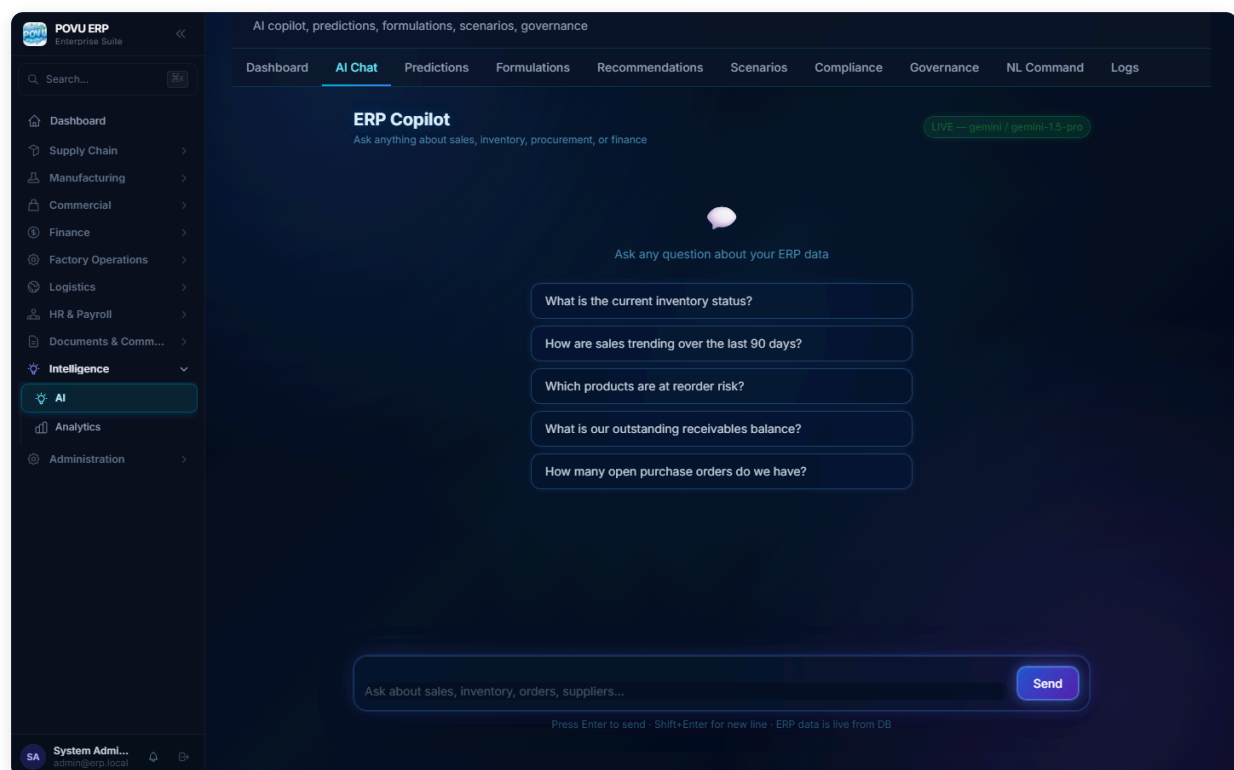
3. Generates SKU-level forecast for configurable horizon
4. Updates weekly (or on-demand)
5. Forecast error tracked (MAPE %)

Using the forecast:

- Sales team: review and adjust forecast in Sales → Forecast
- Planning: MPS uses forecast as input automatically if integrated
- Procurement: reorder suggestions incorporate forecast

9. AI Chat Interface

Route: `/dashboard/ai/chat`



AI chat — natural language queries about your ERP data.

What it does

Ask questions about ERP data in plain English (or Swahili). AI queries the database and returns answers with charts or tables.

Example Queries

Sales:

- "What was our total revenue in April 2026?"
- "Show me the top 5 customers by sales value this year"

- "Which products haven't sold in the last 60 days?"
- "What is the average order value this month vs. last month?"

Inventory:

- "What items are below reorder level right now?"
- "Show me all products expiring within 30 days"
- "What is the total value of finished goods stock?"
- "Which warehouse has the most dead stock?"

Production:

- "What was Line 2 OEE this week?"
- "How many work orders are overdue?"
- "What is the yield % for product SKU-001 this month?"
- "Which machine had the most downtime in May?"

Finance:

- "What is our current accounts receivable total?"
- "Show overdue invoices older than 60 days"
- "What is the budget variance for marketing this quarter?"

HR:

- "How many employees are on leave today?"
- "List employees whose probation ends this month"
- "What is the average attendance rate for the factory this week?"

AI Chat Limitations

- AI reads data only — it cannot create, edit, or delete records
 - Results are as current as the last data sync (real-time for most tables)
 - Complex cross-module queries may take a few seconds
 - If a query returns no results, rephrase or check filters
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10. Common Mistakes & Troubleshooting

Problem	Cause	Fix
Dashboard shows no data	Date filter too narrow	Expand date range; check if data exists for period
Sales chart missing products	Products not linked to correct category	Update product category in Products master
Production analytics wrong	Work orders not closed	Close completed work orders in Production module
Report builder export fails	Too many rows	Add more filters to reduce result set; export max 50,000 rows
AI forecast not updating	Model hasn't run yet	AI forecast updates weekly; or trigger manual refresh
AI chat says "no data found"	Query too specific or table empty	Rephrase; check the underlying module has records
Scheduled report not arriving	Email address wrong	Check recipient email in report schedule settings

11. Related Modules

Analytics Topic	Source Module
Sales revenue data	Sales → Invoices
Production OEE	Manufacturing → Production
Inventory stock	Inventory → Stock
Finance P&L	Finance → Accounting
HR headcount	HR → Employees
Procurement performance	Procurement → Purchase Orders
Quality defect rate	Quality → Inspections
Maintenance MTBF	Maintenance → Breakdowns